Training dynamics of tree tensor networks: summary of findings

Short version of the full report (`report/main.pdf`)

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Setting. A tree tensor network (TTN) represents a tensor W by K small tensors (*cores*) contracted along a tree; matrix product states (MPS, a chain), binary trees and hierarchical Tucker (HT) trees are examples. We train $f(x) = \langle W, x_0 \otimes \cdots \otimes x_{N-1} \rangle$ by gradient flow on the exact population loss $\frac{1}{2}\|W - W^*\|^2$ against a teacher W^* that is a sum of components (e.g. products of basis vectors), starting from random cores of scale σ. The model is multilinear of degree K in its parameters, like a deep linear network, but the tree adds structure: a component can differ from what is already learned on only a few input positions (*sites*), which sit on a few cores. Status tags: [proven] theorem; [derived] derivation under stated assumptions; [empirical] measured; [conjecture] conjecture. Tens of thousands of gradient-flow runs on more than ten tree shapes; every experimental claim was independently rechecked, except the last test in §2.3.

1 Headline results

1. **Small initialization gives saddle-to-saddle learning, one component at a time.** Each plateau sits at a critical point with all bond ranks fixed, and each escape raises the rank of every bond on a *route* by one [empirical]. The first plateau lasts $\propto \sigma^{-(K-2)}$, with a prefactor that one integration of a σ-free flow computes *from the initialization before training*; it predicts the first escape time and which component wins (1,847 of 1,848 runs). The standard bound from the initialization norm is low by up to $3 \cdot 10^7$ [proven][derived][empirical].

2. **The tree, not the size of a component, sets the later plateaus ("locality beats weight").** After a component is learned, the next one needs only the k^* cores of the smallest subtree connecting the cores that carry the sites where it differs (its *Steiner route*). Its plateau is absent for $k^* = 1$, $\propto \ln(1/\sigma)$ for $k^* = 2$, and $\propto \sigma^{-(k^*-2)}$ for $k^* \geq 3$ [proven][derived]. Confirmed with exponent $k - 2$ for $k \leq 5$ in about 11,600 runs on nine trees [empirical]. Consequences: a *smaller* component on a shorter route is learned first as $\sigma \to 0$, and the same teacher has different plateau laws on an MPS (logarithmic) and a binary tree (power law) [empirical].

3. **The tree shape decides whether saddles are strict.** For generic (non-orthogonal) teachers, the saddle after the first escape is strict on an MPS (logarithmic escape at the predicted rate) but higher-order on the binary and HT trees tested, whose internal cores carry no sites (13 teachers) [derived][empirical].

4. **Which component comes first.** If two components differ on sites carried by ≥ 3 cores, the winner is a race fixed by the initialization direction, independent of σ (exact at leading order on an MPS [proven]); on two cores learning becomes greedy as $\sigma \to 0$ [derived][empirical].

5. **Failure mode: dead routes.** Under the usual fan-in initialization some routes die (an end core decays to roundoff), stalling 18–80% of runs with $K \geq 4$ at tight bond dimension. A conserved quantity of each route end, computed from the initialization alone, predicts it: 93% of runs predicted

dead stall vs. 1.8% of those predicted alive [derived][empirical]. Gauge-balanced initialization cannot kill leaf ends.

6. **Map of regimes.** As σ grows: saddle-to-saddle $\to$ a rich window $\to$ lazy. A lazy (kernel) regime exists *only* if every bond reaches its cut dimension; below it training stalls on a floor [derived][empirical]. On binary4 both boundaries scale as $\chi^{-1/2}$ [empirical].

7. **Optimizers.** GD keeps the saddle sequence below a sharp step limit $\eta_c = 2/(K s_1^{2-2/K})$ [proven]; SGD never shortens the first plateau on average, lengthens later ones and can stall a run [empirical]; Adam keeps the first component but makes the first escape σ-independent until its ϵ restores the law [derived][empirical]; ALS, DMRG and Riemannian GD have no plateaus [derived].

8. **Real data (MNIST, pre-registered).** The first route on a binary tree is a sibling pair ($k = 3$), and across 27 pixel orders a data-only measure of short-route coupling predicts the final loss (Spearman -0.60, 95% CI $[-0.82, -0.22]$) but not the speed; spatial locality itself is not the cause [empirical].

2 Theoretical results and how new they are

2.1 Genuinely new (no prior statement found)

- **Saddle-order theorem** [proven] (two independent verifications). At a saddle with one learned component j (weight s_j) and a spare bond channel along the route, define a "generalized piggyback" operator $\mathcal{S}$ that couples the residual through small sets of cores. If $\lambda_{\max}(\mathcal{S}) < s_j$, the saddle order (lowest order at which the loss can decrease) is *exactly* k^*. Otherwise it drops to $2\ell^* < k^*$, with an explicit descent curve and a matching level-wise lower bound. The condition always holds for two-component teachers learned in weight order and for $k^* \le 4$ with a definite old block, so the route rule is a theorem there; it first fails at $k^* = 5$ (an MPS6 counterexample with a positive semidefinite Hessian but order 4, verified numerically to $5 \cdot 10^{-4}$). Previously known: the *upper* bound (a descent curve of order equal to the rank-deficient component, Furman et al., 2026) and lower bounds only for deep linear networks (Achour et al., 2024) and one parity saddle. Prior work does not identify that the relevant set is a Steiner subtree, which can be much smaller than the rank-deficient component.

- **The dynamical consequence** [derived][empirical]. Where the order drops, escape is faster: a pre-registered test (about 2,800 runs) measures exponents 2.001 ± 0.001 (order 4) and 3.996–3.998 (order 6) wherever the initialization lies below the predicted crossover, against the route law; mixed above the crossover; controls inconclusive (dead routes). This test was *not* independently verified.

- **Route-end trap criterion** [derived] (verified). Which routes die is decided by a conserved quantity per route end (its sign), so it is a property of the initialization direction; the death probability of an end is an incomplete-beta law (0.39 at an MPS leaf end, 0.5 at the root); a route is dead when two or more ends die (for $k \le 5$); the four-core case reduces to two rotation invariants [proven]. Relates to known shrinkage of dormant weights and imbalance-induced stalls (Furman et al., 2026), but the route-end condition is new.

- **Strictness by tree shape** [proven][derived]: a learned saddle is strict iff a two-core route exists or a piggyback coupling exceeds s_j; non-orthogonal teachers create two-core routes exactly where adjacent cores carry sites (MPS but not binary/HT).

- **Regimes:** lazy only with full bonds; the lazy boundary constant $\sigma_{\text{lazy}} \approx 6.5/\sqrt{\chi}$ (binary4) [derived][empirical]; a μP-type parameterization for TTNs [derived].

2.2 TTN versions of known mechanisms (new form, not new idea)

- First escape $\propto \sigma^{-(K-2)}$ and its per-initialization blow-up time: known for homogeneous models (Kumar and Haupt, 2025; Kunin et al., 2025); new are the tree evaluation and run-by-run tests.
- Escape laws by number of small layers (none / log / power): known abstractly (Rawal and DeWeese, 2026); new is *which* cores are small later (the Steiner route), its coupling and the closed-form head start.
- Race vs. greedy order: analogous to multi-spike tensor PCA (Ben Arous et al., 2025); new is the classification by number of cores and the exact race on MPS.
- GD stability limit and two-cycles (cf. MacDonald et al., 2025); Adam's ϵ law is concurrent work (Sabanayagam and Lucey, 2026).

2.3 Already known (reproduced in our setting)

The binary tree with four sites is a depth-3 deep linear network, so its saddle sequence, greedy limit and strict wrong-order saddles are known (Saxe et al., 2014; Jacot et al., 2022; Achour et al., 2024); balanced KKT escape directions (Kumar and Haupt, 2025); the two-core case (Jiang et al., 2023); low hierarchical-rank bias of hierarchical tensor factorization (Razin et al., 2022).

3 Main caveats

Results are for exact population loss and whitened Gaussian inputs, small trees ($N \leq 16$, mostly $N \leq 8$) and mostly synthetic teachers. The large-K limit of the saddle-to-saddle/rich boundary is open (a limit of 1 is rejected for the plain model, post hoc). The real-data mechanism is a conjecture. Full scope, numbers, error bars and the record of claims withdrawn after review are in the full report.

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A Figures (from the full report)

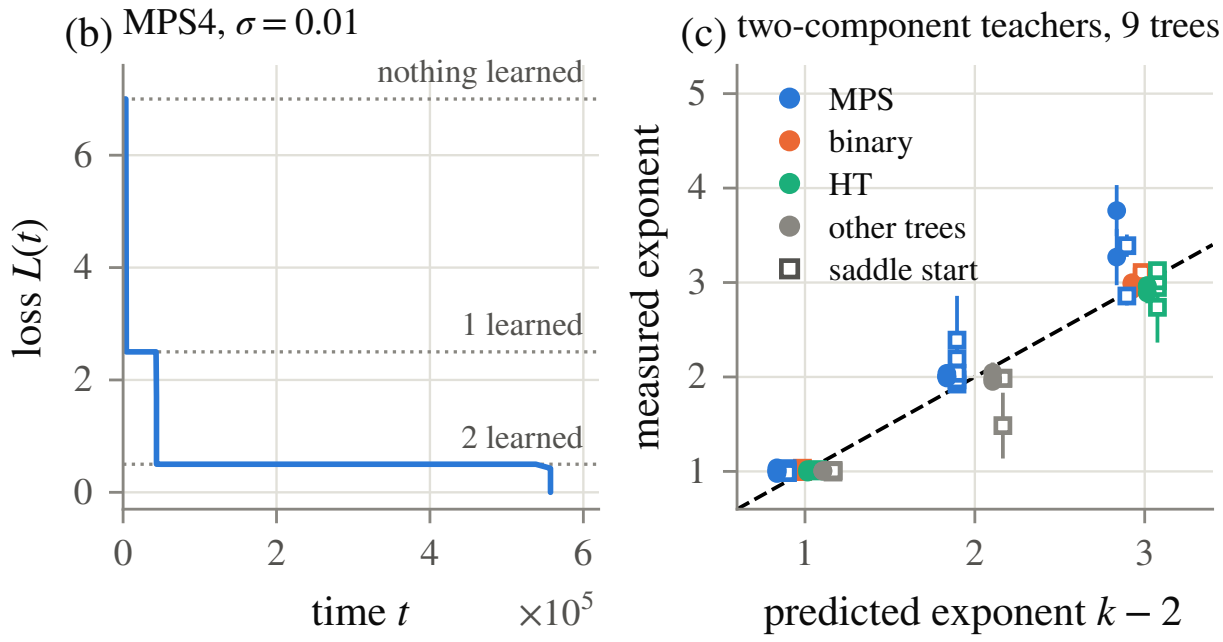


Figure 1: Saddle-to-saddle learning and the route rule (full report, Fig. 1 data panels): loss with partial-fit plateau levels, and the measured plateau exponent against $k-2$.

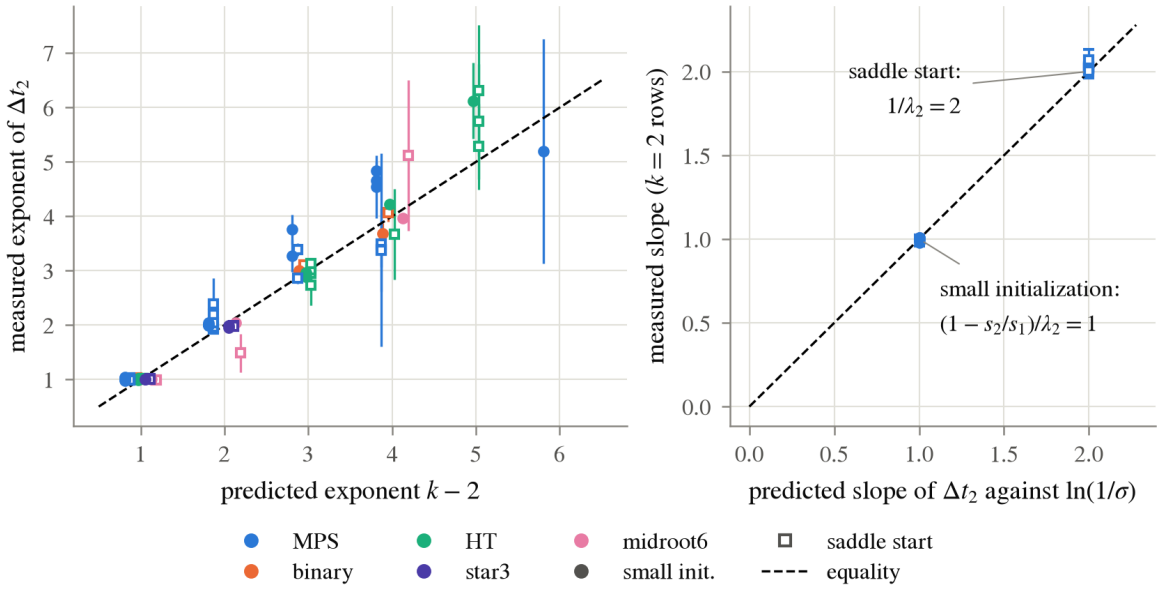


Figure 2: Plateau exponent vs. route size k across nine trees: the measured exponent follows $k-2$ (full report, Sec. 4.4).

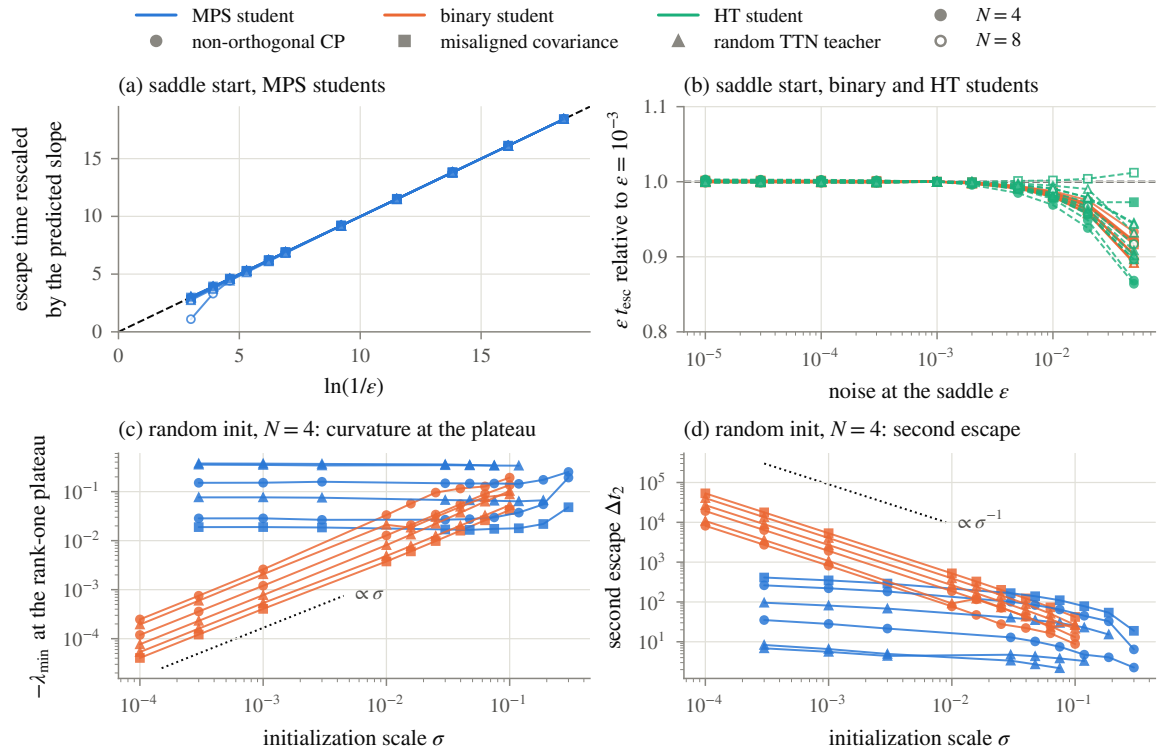


Figure 3: Tree-shape contrast for generic teachers: logarithmic (strict) escape on MPS, power-law (higher-order) on binary/HT trees (full report, Sec. 6.2).

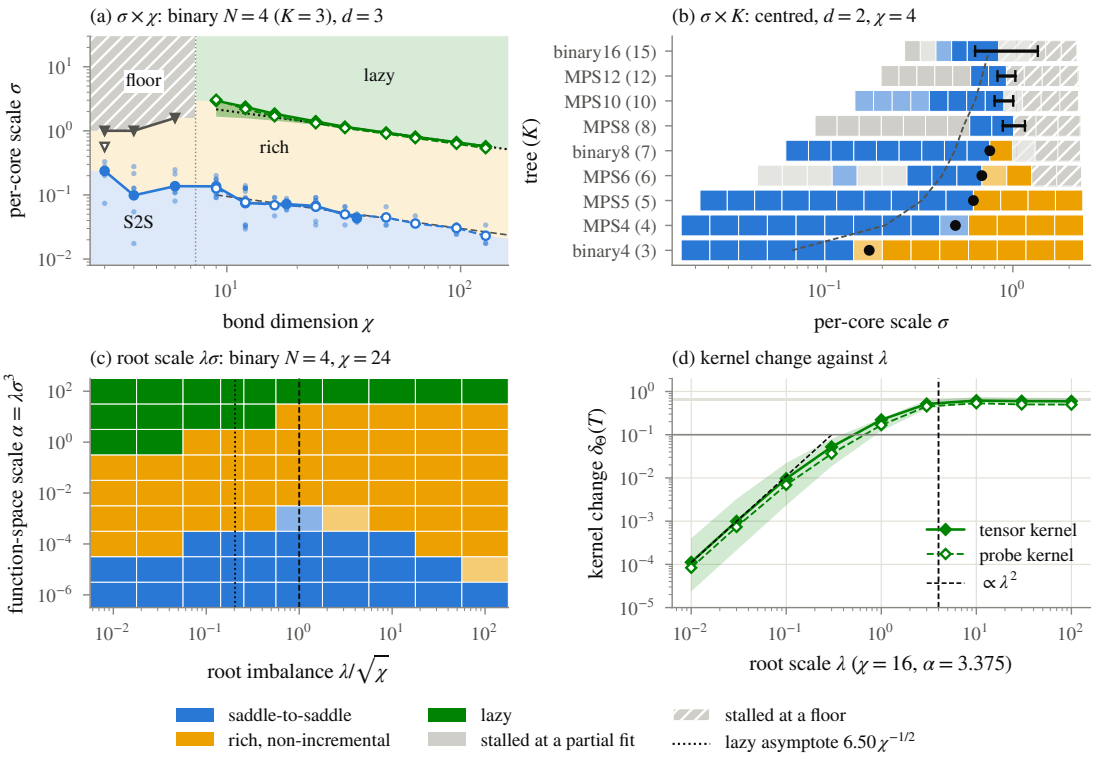


Figure 4: Map of regimes on binary4 in initialization scale σ and bond dimension χ (full report, Sec. 3).